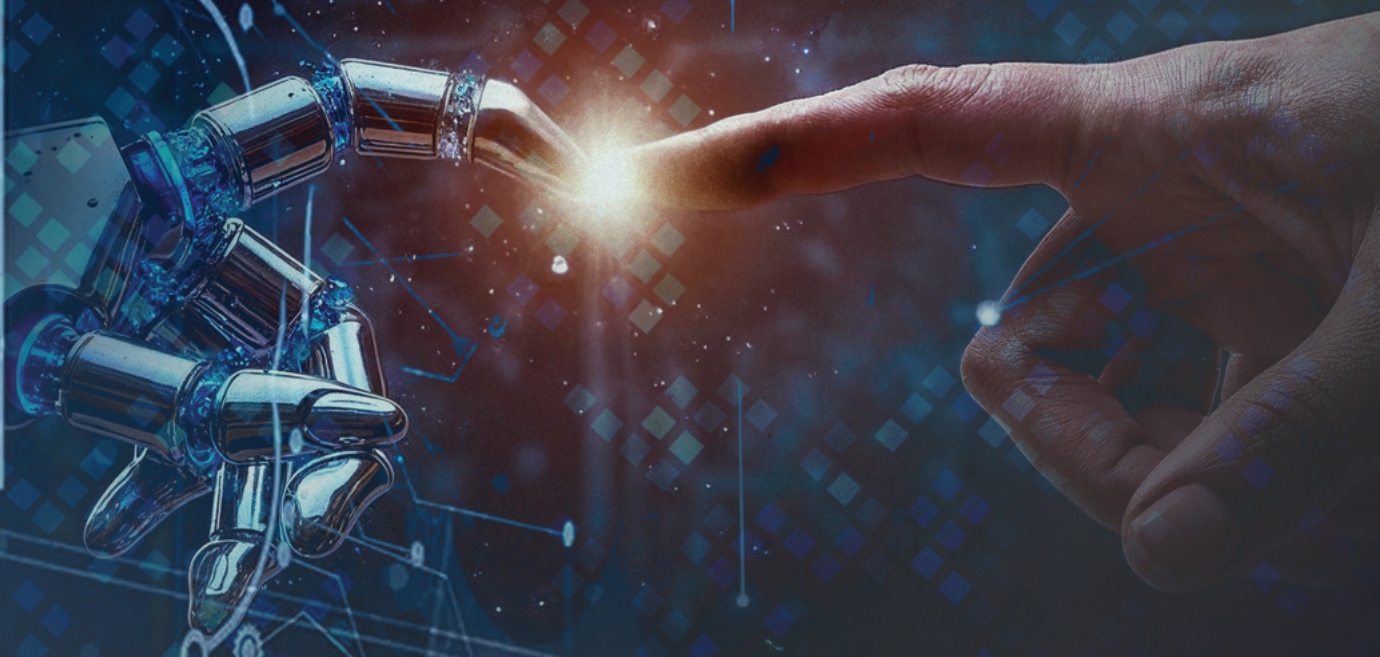


Master Program in **Gen AI**



Stay Ahead
with **FITA**

ACCREDITED BY



Great Future Starts from Here.

A young woman with long dark hair, wearing a light-colored sweater, is sitting at a desk in a classroom. She is raising her right hand, pointing her index finger upwards. She is looking towards the front of the room with a slight smile. In the background, other students are visible, and there are bookshelves filled with books.

ABOUT **FITA ACADEMY**

In 2012, a group of experienced IT leaders from top global MNCs came together with one mission: to turn learning into real career outcomes. What started as a bold idea soon became a clear, structured pathway for anyone aiming to break into the industry. The focus was simple. Delivering skills that companies actually hire for, taught through hands-on training, real projects, and an industry-shaped curriculum.

With our industry-driven curriculum, led by expert trainers, real-time live projects, and flexible online/classroom options that match your pace and proven placement success, we don't just teach; we get our students hired. With unlimited placement support and access to hiring partners, we are building an "industry-ready" workforce.

Transforming Students & IT Professionals into Industry Ready Workforce.



SINCE

2012

LOCATIONS



CHENNAI



COIMBATORE



MADURAI



TRICHY



ERODE



PONDICHERRY



TIRUPPUR

6812

Students Placed Last Year

1000+

Expert Trainers

100%

Placement Support

3000+

Placement Tieups

15+

Branches



Why **FITA Academy**?



Live Capstone Projects



Real time Industry Experts as Trainers



Placement Support till you get your Dream Job offer!



Free Interview Clearing Workshops



Free Resume Preparation & Aptitude Workshops





6,812

**STUDENTS SUCCESSFULLY
PLACED LAST YEAR**

About The Program

Our Gen AI Masters Program prepares you for the next era of applied AI careers. With 14 structured modules, you'll learn how to use prompt engineering, build AI-powered applications, design intelligent assistants, develop agentic workflows, automate business processes, and deploy enterprise-grade solutions on AWS and GCP.

Whether you are a beginner or an experienced professional, the program adapts to your level. Developers gain hands-on experience with LLMs, RAG systems, LangChain, LangGraph, CrewAI/AutoGen, MCP, and cloud AI platforms like Amazon Bedrock, SageMaker, Google Vertex AI, and Gemini. Analysts learn to build data-driven AI workflows using Python and automation tools. Business professionals and entrepreneurs learn to turn ideas into AI products, automate operations, and improve productivity using GenAI tools.

Each module is designed for practical learning and real-world application. The curriculum includes Prompt Engineering, Python for GenAI, LLM architecture, RAG systems, AI agents, LangChain, LangGraph for workflow orchestration, CrewAI/AutoGen for multi-agent systems, MCP, n8n automation, and Vibe Coding using GitHub Copilot, Claude, Codex, Emergent, and Google Antigravity. Learners also gain exposure to embeddings, vector databases, memory systems, tool integrations, and AI deployment practices.

Responsible AI concepts such as hallucination control, bias handling, safety, and data privacy are included throughout the program. The course ends with real-world projects and an AI Hackathon where learners build and deploy complete Generative AI solutions. This program is designed for professionals who want to upgrade skills, automate workflows, and build strong careers in Generative AI.

Software Stack You'll Master in This Program



Key Features of the Program

- Understand the foundations of Generative AI — including the evolution of AI, LLMs, diffusion models, multimodal AI, and real-world applications across industries like healthcare, education, marketing, and finance.
- Master Prompt Engineering techniques using ChatGPT, Gemini, Claude, and other leading AI models for content generation, automation, reasoning, and business workflows.
- Learn Python fundamentals, API integrations, JSON handling, and AI libraries required for building modern Generative AI applications and workflows.
- Understand Large Language Models (LLMs), tokens, context windows, embeddings, inference, fine-tuning, and deployment approaches using GPT, Gemini, LLaMA, and Mistral models.
- Build Retrieval-Augmented Generation (RAG) systems using embeddings, vector databases, semantic search, chunking strategies, and LangChain-powered retrieval pipelines.
- Design and develop advanced AI Agents using LangChain, LangGraph for stateful workflow orchestration, memory systems, tool integrations, multi-step reasoning, and autonomous execution.
- Build multi-agent AI systems using CrewAI/AutoGen to enable role-based agents, collaboration, task delegation, and autonomous problem-solving workflows.
- Gain practical exposure to MCP (Model Context Protocol) and learn how AI models securely interact with tools, APIs, databases, browsers, and external resources.
- Automate business and AI workflows using n8n through triggers, APIs, integrations, conditional logic, scheduling, and intelligent automation pipelines.
- Deploy and manage Generative AI applications using AWS Bedrock, SageMaker, Google Vertex AI, Gemini models, and scalable cloud deployment workflows.
- Learn modern Vibe Coding practices using GitHub Copilot, Claude, Codex, Emergent, and Google Antigravity to generate code, build project scaffolds, automate debugging, and accelerate end-to-end development workflows.
- Explore responsible AI practices, including hallucination mitigation, bias awareness, data privacy, and compliance considerations when building and deploying AI systems.
- Participate in a final AI Hackathon to design and build complete end-to-end Generative AI solutions using modern frameworks, cloud platforms, and AI development tools.

Top High-Paying Career Roles You Can Land

Generative AI Engineer

AI Agent Developer

RAG Pipeline Engineer

AI Automation Specialist

No-Code AI Developer

AI Business Analyst

Prompt Engineer

Conversational AI Designer

Vibe Coder

**AI Quality Assurance (QA)
Tester**



Who is This Program Ideal For

1

Freshers & Graduates

Go beyond basic theory, gain practical, beginner-friendly AI skills, and build your portfolio with real-world projects that help you land your first high-growth role in the advanced GenAI industry.

2

Working & IT Professionals

Future-proof your career and boost your earning potential by applying AI directly to your existing role.

3

Career Transitioners

Go beyond theory, gain beginner-friendly AI skills, and build a portfolio with real-world projects to land your first high-growth GenAI role. Combine your existing domain knowledge with GenAI skills to deliver high-value solutions or move into lucrative roles.

The background of the entire page is a blue-tinted image. It features a person from the chest up, wearing a VR headset. Overlaid on the person and the background is a complex, glowing digital circuit or network pattern, suggesting a high-tech or artificial intelligence theme.

4

Entrepreneurs

Cut operational costs, automate repetitive tasks, multiply output without hiring more people, and build a leaner, faster business with AI.

5

Career Gap Professionals

Re-enter the workforce with updated, in-demand, job-ready AI skills that open flexible, well-paid opportunities in AI operations and digital transformation.

INDUSTRY TRENDS



9/10

Software engineers lack production-ready GenAI skills.

Source: LinkedIn / NASSCOM insights



\$120K–\$200K+

This is the average salary range for skilled GenAI professionals globally.

Source: Glassdoor



40%

Productivity improvement is seen in content creation using GenAI tools.

Source: McKinsey Digital



70–80%

Of enterprises are actively adopting or planning Generative AI solutions.

Source: Gartner / IDC

Master Program in Gen AI

Module 1: Introduction of GenAI

- Overview of Generative AI
- Evolution of AI: From rule-based to generative models
- Key concepts: LLMs, diffusion models, multimodal AI
- Real-world applications across industries
(education, marketing, healthcare, finance)
- Career paths in Generative AI

Module 2: Prompt Engineering using ChatGPT & Gemini

Fundamentals of Prompt Creation

- Anatomy of a prompt (instructions, context, constraints)
- Prompt elements (roles, input, output format)
- Zero-shot, one-shot, and few-shot prompting
- Structured vs unstructured prompts

GEN AI

Techniques for Effective Prompts

- Chain-of-thought prompting
- Role prompting, persona-based prompts
- Tree of Thoughts (ToT)
- Directional stimulus prompting

Prompt Tuning for AI Models

- Using system vs user prompts
- Refining prompts for accuracy, creativity, and consistency
- Evaluating prompt effectiveness

Case Studies and Hands-On

- Business use cases: Email writing, marketing copy, brainstorming
- Creative use cases: Storytelling, poetry, art descriptions
- Code Generation Use Cases
- Exercises: Zero vs few-shot comparison, role-based Chatbot, prompt chaining workflow

Module 3: Python for GenAI

Introduction to Python & Setup

- Why Python is important for GenAI
- Installing Python / Using Google Colab or Jupyter Notebook
- Basic IDE navigation

Python Fundamentals

- Variables and data types (string, integer, float, boolean)
- Basic operators (+, -, *, /, %, //, **)
- Input and output (print(), input())

Working with Data

- Lists, tuples, dictionaries, sets
- Indexing, slicing
- Adding, updating, and removing data
- Looping through data (for, while)
- Basic list comprehensions

Functions & Modular Code

- Defining functions (def)
- Function parameters and return values
- Built-in functions vs custom functions
- Importing Python modules

Working with Files

- Reading and writing text files
- JSON basics (important for API data exchange)
- Loading & saving datasets

API Interaction

- Introduction to APIs in Python
- Using the requests library to call an API
- Sending parameters and receiving responses
- Parsing JSON responses



Working with Libraries for GenAI

- Introduction to popular AI-related Python libraries:
- OpenAI (OpenAI API)
- pandas (data handling)
- numpy (numerical data)
- JSON (data exchange)
- Examples of calling a GenAI model Project: Create a script that:
 - Reads a prompt from a file
 - Sends it to a GenAI API (ChatGPT/Gemini/Claude)
 - Saves the response to a file
- Discussion of results and improvements
- Best practices for Python in GenAI
- How to apply Python in real GenAI workflows

Module 4: Large Language Models (LLMs) – Architecture, APIs, and Deployment

Introduction to LLMs

- What are LLMs and why do they matter
- Examples: GPT, Gemini, LLaMA

Architecture & Key Concepts

- Overview of GPT, Gemini, and LLaMA
- Tokens, context length, inference latency, model size
- Strengths and limitations of each LLM

Deployment Approaches

- On-prem vs Cloud LLMs
- Deployment choices, trade-offs, and cost factors

Hands-On with APIs

- OpenAI ChatGPT API (text generation, embeddings)
- Google Gemini API (multimodal inputs)
- Meta LLaMA API (open-source customization)
- Running LLMs on-prem (LLaMA, Mistral)

Parameter Tuning & Customization

- Fine-tuning vs parameter-efficient tuning (LoRA, PEFT, adapters)
- Prompt-tuning vs instruction-tuning
- Safety alignment & RLHF
- Best practices for domain-specific performance

Ethical Considerations

- Hallucination risks and mitigation
- Bias & fairness
- Data privacy and compliance

Project: Build a Domain-Specific LLM-Powered Intelligent Assistant

- Education Assistant: Helps students with queries, creates study plans, and summarizes lessons
- Customer Support Chatbot: Handles FAQs, routes complex issues, and generates ticket summaries
- Business Knowledge Assistant: Answers queries on company data, generates sales pitches, and produces reports
- Creative Writing Assistant: Generates story ideas, creates marketing copy, or writes poetry
- Healthcare Q&A: Answers medical FAQs with disclaimers, generates summaries of research

Students can pick any one project from the list above.

Module 5: Basics of RAG

Introduction to RAG

- Why LLMs need external knowledge
- How RAG reduces hallucinations
- Real-world use cases

Embeddings Basics

- Text vector meaning representation
- Semantic similarity concept
- Role of embeddings in retrieval

Chunking & Document Preparation

- Splitting documents into chunks
- Chunk size & overlap
- Cleaning/preprocessing text

Vector Databases

- Purpose: store & search embeddings
- Why SQL/NoSQL won't work
- ChromaDB



Retrieval Methods

- Similarity search
- MMR (relevance + diversity)
- Hybrid search

RAG Pipeline Overview

- Query embedding search context answer
- How retrieved data improves responses
- Architecture flow

No-Code RAG Implementation

- Upload docs (LlamaIndex/Lamini)
- Auto-generate embeddings
- Test retrieval & context

Best Practices & Common Pitfalls

- Clean documents & good chunking
- Use strong embedding models
- Avoid noisy or irrelevant data



Module 6: Advanced AI Agents & RAG

Building using LangChain

Introduction to LangChain & Concepts

- Overview of LangChain architecture and ecosystem
- Why LangChain is powerful for AI agents and RAG
- Key concepts:
 - Chains
 - Agents
 - Tools
 - Memory
 - Prompt templates
- How LangChain supports multi-step reasoning and retrieval augmentation
- Setting up a LangChain environment
- Overview of LangChain APIs and SDK
- Building a basic chain
- Running a simple prompt using LangChain

Building RAG Pipelines in LangChain

- Ingest documents into the LangChain retriever
- Chunking & embedding generation
- Storing embeddings in the retriever
- Building a retrieval chain
- Injecting retrieved context into prompts
- Testing and optimizing retrieval-based answers

A graphic of a glowing blue circuit board with a central square component labeled "AI Agent" in white text. The background is dark blue with faint circuit patterns and a glowing blue sphere at the bottom right.

AI
Agent

Building Advanced AI Agents in LangChain

- Understanding LangChain agents
- Zero-shot and multi-step reasoning agents
- Memory integration
- Tool usage in agents
- Creating a LangChain agent
- Adding memory for context retention
- Connecting APIs as tools
- Building a multi-step workflow
- Example project: A conversational agent that performs reasoning with retrieval and actions

Project: LangChain AI Agent with RAG

- Project goal: Build an advanced domain-specific AI assistant with retrieval and multi-step reasoning

Steps:

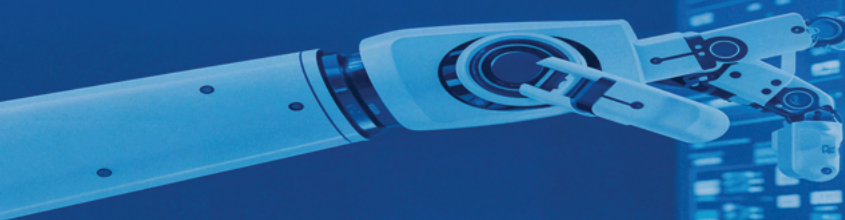
- Define the problem statement and scope
- Implement a retrieval pipeline in LangChain
- Build an agent that uses retrieval output and memory for multi-step reasoning
- Add tool integrations (API calls, web scraping, etc.)
- Test and optimize performance
- Building Advanced AI Agents in LangChain
- Deliverable: Fully functional LangChain AI Agent with RAG and memory capabilities

Module 7: Building Stateful AI Workflows using LangGraph

- Introduction to LangGraph
- Graph-based agent architecture
- State management
- Human-in-the-loop systems
- Multi-step workflows
- Agent orchestration patterns
- Advanced workflow design
- LangGraph project

Module 8: Multi-Agent Systems using CrewAI / AutoGen

- Agent collaboration
- Roles and tasks
- Agent communication
- Delegation workflows
- Autonomous task execution
- Multi-agent architectures
- CrewAI/AutoGen implementation
- Multi-agent project



Module 9: Introduction to MCP (Model Context Protocol)

Introduction to MCP

- What is Model Context Protocol (MCP)
- Why is MCP important for AI applications
- How MCP connects AI models with tools and data

Understanding MCP Architecture

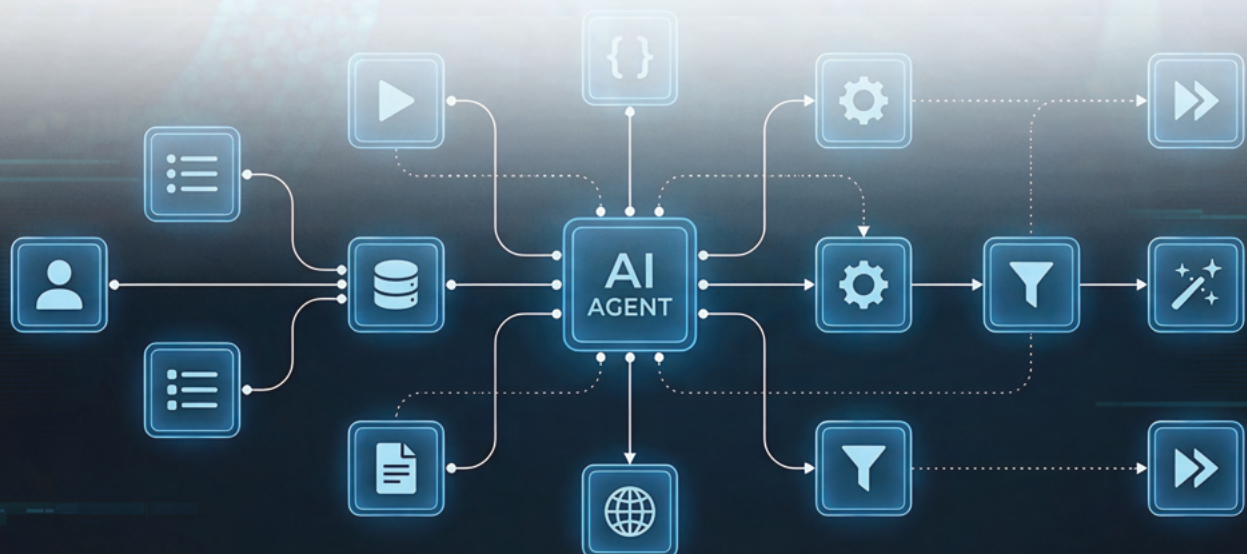
- Client, Server, and Host concepts
- Communication flow in MCP
- How models access external resources safely

MCP Tools & Resources

- Using tools through MCP
- Accessing files, APIs, databases, and browsers
- Resource handling and permissions

MCP Workflow Basics

- Request and response flow
- Tool calling process
- Context sharing between applications and models



MCP Use Cases

- AI assistants with external tools
- IDE and coding assistant integrations
- Business automation and workflow support

Hands-on MCP Setup

- Installing MCP-compatible tools
- Connecting models with MCP servers
- Basic configuration and testing

Building Simple MCP Applications

- Creating a basic MCP server
- Exposing tools to AI models
- Testing simple prompts with tool usage

Security & Best Practices

- Permission control and safe access
- Managing sensitive data
- Reducing unnecessary tool exposure
- Best practices for scalable MCP workflows

Future of MCP

- Growing adoption in AI ecosystems
- MCP in enterprise AI applications
- Role of MCP in autonomous AI agents



Module 10: Vibe Coding using (GitHub Copilot / Claude / Codex)

Introduction to Vibe Coding

- What is Vibe Coding
- Evolution of AI-assisted software development
- Overview of GitHub Copilot, Claude, Codex, Gemini & other AI coding tools

AI-Powered Code Generation

- Crafting effective prompts for accurate code generation
- Generating CRUD operations, APIs, middleware, config files, and project scaffolds
- Using AI tools for frontend and backend development
- Working with reusable templates and boilerplates

Code Explanation & Reverse Engineering

- Understanding complex functions using natural language explanations
- Analyzing architecture, workflows, and design patterns with AI assistance
- Identifying code smells and refactoring suggestions
- Using AI copilots to understand legacy or large-scale projects

Debugging & Code Optimization

- AI-assisted debugging and error handling
- Understanding stack traces and resolving compilation/runtime issues
- Optimizing API calls, queries, and application performance
- Performance tuning using AI suggestions

**VIBE
CODING**

Future of AI-Assisted Development

- AI copilots in modern software engineering
- Role of AI in rapid prototyping and automation
- Trends in autonomous and agentic coding systems

Module 11: Vibe Coding using Emergent

Emergent Platform Mastery

- Setting up and navigating the Emergent platform environment.
- Understanding the platform's core engine and how it interprets high-level developer intent.

Rapid Prototype Development

- Building functional application scaffolds from a single natural language prompt.
- Iterative UI/UX design through conversational feedback loops.

Multi-Agent Orchestration & Refinement

- Coordinating specialized agents for frontend, backend, and logic within Emergent.
- Real-time previewing, reviewing, and refining visual artifacts.

Emergent-to-Production Workflows

- Bridging the gap between conceptual intent and deployable codebases.
- Integrating Emergent workflows with traditional development pipelines and local environments.



Module 12: Vibe Coding using Google AntigraVity

Agent-First Architecture

- Introduction to Agent-First Development
- Evolution from Traditional IDEs to Agent Orchestration
- Principles of Agent-Driven Platforms

Antigravity Workflow & VS Code Fork

- Antigravity VS Code Setup
- Editor Interface & Managerial View
- Mission Control & Task Spawning
- Goal-Based Prompt Writing
- Multi-Agent Coordination
- End-to-End Workflow Execution

Practical Agentic Use Cases

- Building Apps from a Single Prompt
- UI Iteration using Vibe Coding
- Reviewing Visual Artifacts
- Automated Code Refactoring
- Bug Detection & Debugging
- Automated Testing



Module 13: Building Advanced AI Agents using n8n

Introduction to n8n

- What is n8n and why it is important
- Key features and benefits
- n8n vs other automation tools (Zapier, etc.)
- Core concepts: workflows, nodes, triggers, actions, credentials

Getting Started with n8n

- Setting up n8n (cloud vs self-hosted)
- Navigating the n8n UI
- Understanding the workflow editor
- Understanding node types: trigger nodes, action nodes, function nodes, code nodes

Basic Workflows in n8n

- Creating your first workflow
- Using triggers (manual, webhook, schedule)
- Adding action nodes for APIs and services
- Connecting nodes and handling data flow
- Testing and executing workflows

Advanced Workflow Features

- Using variables and expressions in workflows
- Using conditional logic in workflows
- Error handling and workflow retries
- Data transformation with function and code nodes
- Looping and iteration in workflows

Integrations with n8n

- Connecting n8n to APIs and external services
- Popular integrations: Google Sheets, Gmail, Airtable, Notion, etc.
- Creating custom integrations via API requests, using authentication and credentials securely

n8n for Automation

- Building automation workflows for business use cases:
 - Automated email responses
 - CRM automation
 - Data syncing between apps
 - Report generation and distribution
- Scheduling workflows for periodic tasks

Building an n8n Project

- Automated lead generation workflow
- Document-based knowledge assistant
- Social media posting automation
- Define the use case and scope
- Design the workflow
- Configure nodes and triggers
- Test and debug workflows
- Optimize workflow execution

Best Practices

- Workflow structuring and naming conventions
- Version control and backup of workflows
- Security best practices for credentials
- Monitoring and maintaining workflows

Module 14: Cloud Deployment for Generative AI (AWS / GCP)

AWS Generative AI Deployment

- Why AWS for GenAI
- Amazon Bedrock
- Foundation Models in Bedrock
- Bedrock Playground Demo
- Amazon SageMaker Basics
- Deploying Pre-trained Models
- Scaling and Monitoring

GCP Generative AI Deployment

- Why GCP for GenAI
- Vertex AI Overview
- Gemini Models in Vertex AI
- Vertex AI Studio
- Model Garden
- Deploying and Testing AI Models
- Monitoring and Cost Management

AI Hackathon

300+ CORPORATE CLIENTS

amazon

cognizant

DELL

VOLVO

epi

ti. HEXAWARE

DEMAND

minuscule
TECHNOLOGIES

2adpro

Ignitho

Autointelli®
Aggregate | Automate | Audit

JEEVAN
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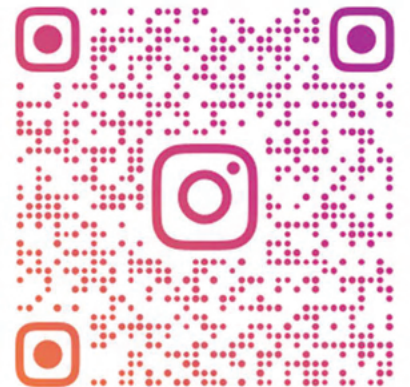
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